DEPARTMENT OF THE AIR FORCE
Headquarters US Air Force
Washington, D.C. 20330-1030

QTP24-3-B230
8 October 2026

OVERSIZED CARGO TRUCK

Vehicle Management Codes: B230-B232, B233-B235, B237, B239-B241, B255, B265-B267, B274, B313, C181, C192, C198, C200, C203-C205, C207, C209, C211



QUALIFICATION TRAINING PACKAGE

Chapter 1—OVERVIEW **4**
1.1. Overview 4
1.2. Administrative 4
Chapter 2—RESPONSIBILITIES AND PERPARATION **5**
2.1. Trainer 5
2.2. Trainee 5
2.3. Certifier 5
Chapter 3—INTRODUCTION **6**
3.1. Objectives 6
3.2. Desired Learning Outcomes 6
Figure 3.1. Recommended Training Time for Training Activities 6
3.3. Lesson Duration 6
3.4. Instructional References 6
3.5. Instructional Training Aids and Equipment 7
Chapter 4—KNOWLEDGE OVERVIEW AND OPERATIONAL REQUIREMENTS **8**
4.1. Overview of Training and Requirements 8
4.2. Vehicle Inspection Procedures 8
4.3. Vehicle Safety, Equipment, and Precautions 10
4.4. Vehicle Operation 10
Chapter 5—AIR BRAKES **12**
5.1. Safe Operation of a CMV with an Air Brake System 12
Figure 5.1. Air Brake System Components 12
Figure 5.2. Manual Drain Valve 13
Figure 5.3. S-cam Air Brakes 14
5.2. Using Air Brakes 17
Chapter 6—EXPLANATION AND DEMONSTRATION **20**
6.1. Trainer Preparation and Safety Protocols 20
6.2. Operator Maintenance and Vehicle Pre-Trip Inspection 20
6.3. Operational Demonstration 20
Chapter 7—TRAINEE PERFORMANCE AND EVALUATION **22**
7.1. Trainee Performance and Practical Application 22
7.2. Trainee Vehicle Inspection Execution 22
7.3. Trainee Operational Execution 22

7.4. Performance Evaluation 22
Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION 23
Attachment 2—OVERSIZED CARGO TRUCK INSPECTION GUIDE 25
Attachment 3—PERFORMANCE TEST 26
Attachment 4—SEVEN-STEP INSPECTION GUIDE 28

Chapter 1

OVERVIEW

1.1. Overview. This Qualification Training Package (QTP) establishes standardized training requirements, content, and certification protocols for all Air Force and Space Force operators of this vehicle. Designed to ensure consistency and excellence across all installations, this package provides comprehensive guidance for trainers, trainees, and certifiers throughout the certification process.

1.1.1. Instructional Standards. The training standards outlined herein align with Department of the Air Force (DAF) requirements and serve as the authoritative source for curriculum development and performance evaluation. Trainers will utilize Chapters 2 through 7 to guide trainee preparation, conduct classroom lectures, provide demonstrations, and administer knowledge and performance evaluations.

1.1.2. Commercial Driver's License (CDL) Integration (Air Brakes Only). If the vehicle being operated is equipped with an air brake system (ABS), the Commercial Driver's License Manual, American Association of Motor Vehicle Administrators (AAMVA) will supplement the baseline requirements found within this QTP. For ease of access, this supplementary air brakes information has been integrated into **Chapter 5**. If the vehicle is not equipped with an air brake system, review of the CDL Manual, **Section 5 Air Brakes** is not required.

1.2. Administrative. Send comments and suggested improvements on DAF Form 847, *Recommendation for Change of Publication*, through Air Force Installation and Mission Support Center (AFIMSC) functional managers via e-mail at: AFIMSC.A34OG.GroundTrans@us.af.mil.

Chapter 2

RESPONSIBILITIES AND PREPARATION

2.1. Trainer. The trainer is responsible for the overall execution of the training plan. This includes providing an overview of **Chapters 2** and **3**, delivering classroom lectures based on **Chapter 4** and **5** (if applicable), and conducting knowledge training and demonstrations (**Chapter 6**). The trainer will administer and grade all review questions, ensuring that any missed items are thoroughly reviewed with the trainee to guarantee task comprehension before signing off on the training block.

2.2. Trainee. The trainee must actively participate in the QTP process and meet all preparation requirements before beginning practical instruction.

2.2.1. Administrative Requirements: The trainee must possess a valid state driver's license and ensure Sections I and II of the DAF Form 171, *Request for Driver's Training and Addition to U.S. Government Driver's License,* are completed in accordance with (IAW) DAFI 24-301, *Ground Transportation.* Additionally, all training and licensing must comply with any applicable local licensing jurisdiction requirements.

2.2.2. Mandatory Reading: The trainee must read this entire QTP and the manufacturer's operator's manual for the specific vehicle being trained on.

2.2.3. Air Brake Exception: If the vehicle is equipped with air brakes, the trainee is additionally responsible for reading **Chapter 5**. This information was transcribed from the CDL Manual (AAMVA).

2.2.4. Execution: During instruction, the trainee will follow along with the lecture, utilize **Attachments 2** and **4** as inspection guides, and prepare for the unassisted performance test.

2.3. Certifier. The certifier acts as the independent evaluating authority. They are responsible for evaluating the Airman's physical task performance (**Chapter 7**) without providing any assistance or coaching. Upon successful, independent completion of the performance test, the certifier will add their name, date, and signature, to sign off on the task(s) finalizing the licensing process.

Chapter 3

INTRODUCTION

3.1. Objectives. The objective of this course is to ensure personnel become fully qualified oversized cargo truck operators who possess the requisite knowledge and skills for safe, professional execution. Through a comprehensive curriculum of lectures, demonstrations, and hands-on driving sessions, trainees will be trained in both the safe operation and preventive maintenance of various oversize cargo truck platforms. Ultimately, trainees must demonstrate their proficiency by independently performing operator inspections and completing the final performance test (Go/No-Go checklist) with zero instructor assists.

3.2. Desired Learning Outcomes.

3.2.1. Understand the safety precautions to be followed pre-, during-, and post- operation of the oversized cargo truck.

3.2.2. Understand the purpose of the oversized cargo truck and its role in the mission.

3.2.3. Know the proper operator maintenance procedures of the oversized cargo truck, IAW applicable technical orders (TOs) and use of DAF Form 1800, *Operators Inspection Guide and Trouble Report*.

3.2.4. Safely and proficiently operate the oversized cargo truck.

3.3. Lesson Duration. Recommended instructional and hands on training time is 23 hours for most trainees (time may vary depending on member).

Table 3.1. Recommended Training Time for Training Activities. Note: The durations listed in Table 3.1 represent recommended baseline training times and are not absolute mandates. Actual training durations may be adjusted at instructor discretion to be shorter or longer, depending entirely on the trainee's progress and demonstrated task proficiency.

Training Activity	Training Time
Trainee's Preparation	2 Hours
Instructor's Lecture and Demonstration	5 Hours
Trainee's Personal Experience (to build confidence and proficiency) ▪ Perform Operator Maintenance ▪ Operate the Vehicle	14 Hours
Trainee's Performance Evaluation	2 Hours

3.4. Instructional References.

3.4.1. Risk Management (RM) and Safety Principles.

3.4.2. Applicable TOs or Manufacturer's Operator's Manual (see Vehicle Management for TO number for vehicle being used in training).

3.4.3. DAF Form 1800.

3.4.4. The CDL Manual for air brake equipped vehicles only.

3.5. Instructional Training Aids and Equipment.

3.5.1. Oversized Cargo Truck QTP.

3.5.2. Oversized cargo truck.

3.5.3. Applicable TO or manufacturer's operator's manual.

3.5.4. DAF Form 1800.

3.5.5. Suitable training area.

3.5.6. Traffic cones or suitable markers.

Chapter 4

KNOWLEDGE AND OPERATOR PERFORMANCE

4.1. Overview of Training and Requirements. The primary learning outcome of this section is for the trainee to understand the safety precautions required pre-, during, and post-operation of oversized cargo trucks. Trainees must also understand the purpose of the vehicle, which varies based on its specific type (e.g., cargo movement, passenger movement, emergency services support), and its role in supporting the unit, base, and overall Air Force mission.

4.1.1. Oversized Cargo Truck Design. The design of an oversized cargo truck varies significantly depending on the vehicle type. Operators must refer to the manufacturer's operator's manual for specific information and to the vehicle data plate for safe load capacity guidance. Generally, oversized cargo trucks feature 2 to 3 axles (one front steer, and one to two rear dual-wheel drives), 6 to 10 total wheels, manual transmissions, and air brakes.

4.1.2. The Four-Wheel Drive (4WD) System. Many oversized cargo trucks are equipped with a 4WD system, allowing the vehicle to supply power to all four wheels simultaneously via a transfer case or power transfer unit. This increased traction enables safe operation over terrain and road conditions that conventional two-wheel drive vehicles cannot navigate. Operators must note that an initial shift from two-wheel drive to 4WD while moving may cause a momentary clunking or ratcheting sound; this is a normal mechanical function as the front drivetrain comes up to speed.

4.2. Vehicle Inspection. Operators must conduct pre-trip, during-use, and post-trip vehicle inspections utilizing the Seven-Step Inspection Method (**Attachment 4**) and the walk-around guide (**Attachment 2**). Any discrepancies found must be reported immediately to the Vehicle Control Official (VCO), the supervisor, or vehicle maintenance. Note that vehicle maintenance may authorize continued use for minor discrepancies that do not pose a safety or breakdown hazard.

4.2.1. Pre-Trip Inspection Checkpoints. During the pre-trip inspection, the operator is looking for specific indicators of damage, wear, or missing items across all major vehicle systems.

4.2.1.1. Engine Compartment & Fluids: Check fluid levels (engine oil, coolant, power steering fluid, transmission fluid, and antifreeze). Inspect the battery for security, fluid level, damage, and corrosion. Check drive belts for tension and fraying and visually inspect all hoses and wiring.

4.2.1.2. Exterior & Undercarriage: Inspect the overall vehicle for cleanliness, damage, or missing items. Look for fuel, oil, coolant, hydraulic, or air leaks under the transmission, differential, shocks, and brakes. Ensure the exhaust system (pipes, mufflers, vertical stacks) is secure with no missing clamps or brackets, has no leaks, and is not rubbing against fuel systems or tires. Verify the fuel door and cap are intact and not damaged, hood latches are secure, and there are no loose, cracked, broken, or missing frame members.

4.2.1.3. Tires, Wheels, and Rims: Verify proper inflation. **Note:** Notify the VCO,

supervisor, and/or vehicle maintenance immediately if a split rim is completely flat. Inspect sidewalls and tread depth, looking for bulges, cuts, abrasions, or tread separation. Ensure valve stems are not cut or cracked. Dual tires must not touch each other or vehicle parts. Ensure tires are not mismatched sizes, radial and bias-ply tires are not mixed, and remove large rocks stuck between wheels. Re-grooved, recapped, or retreaded tires on the front wheels are prohibited. Check all wheel rims for cracks, splits, or welding repairs (which are unsafe). Check for loose/missing lug nuts (rust around nuts indicates looseness), and ensure clamps, spacers, studs, and lock rings are not missing, mismatched, bent, or cracked.

4.2.1.4. Brakes, Steering, and Suspension: Check brake drums for cracks; ensure brake shoes/pads are not dangerously thin, missing, broken, or contaminated with oil, grease, or brake fluid. Inspect the steering system (steering column, gear box, tie rods) for bent, loose, or missing parts (nuts/bolts/cotter keys) and ensure steering wheel play does not exceed 10 degrees (approx. two inches on a 20-inch wheel). Check the suspension for leaking shock absorbers, damaged/leaking air systems, cracked/broken/shifted spring leaves, and secure torque rods, U-bolts, and spring hangers.

4.2.1.5. Cab Interior & Accessories: Check the control panel, heater/defroster, horn operation, and ensure all seatbelts function properly. Inspect the windshield, windows, doors, mirrors, and windshield wipers/washers for damage. Verify interior and exterior wiring, lights, and reflectors are working.

4.2.1.6. Emergency Equipment & Cargo: Confirm the presence of spare electrical fuses, warning devices, a spare tire/jack/lug wrench, and a fire extinguisher. Note: Additional external fire extinguishers are required when transporting some hazardous materials. Ensure cargo is balanced, secured, and not overloaded. If transporting hazardous material (HAZMAT), verify proper labels, placards, and paperwork.

4.2.2. During Operation (Enroute) Inspection. While operating the vehicle, the operator must maintain continuous situational awareness to identify potential mechanical issues or cargo shifting. This includes constantly monitoring the instrument panel and gauges for abnormal pressures, temperatures, or erratic movements. The operator must pay close attention to any unusual sounds or vibrations originating from the cab, undercarriage, or cargo area. When applying the brakes, particularly on downgrades, the operator should utilize their mirrors to watch the tire and brake areas for smoke or other signs of overheating. Additionally, if operating a flatbed cargo truck, the operator must frequently check their mirrors to keep a constant eye on the cargo, ensuring that all tie-downs, straps, and the load itself remain fully secure and balanced throughout the transport

4.2.3. Post-Trip Inspection. Following operation, the operator must check all gauges (oil pressure, fuel, water temperature, voltage) and warning lights/indicators for proper operation. Listen for unusual exhaust or air leaks, and remain alert for abnormal odors, handling problems, or vibrations. The vehicle must be cleaned, all equipment properly stowed, refueled, and safely parked with the brakes applied and transmission in neutral (or park for an automatic).

4.3. Vehicle Safety, Equipment, and Precautions. Operating an oversized cargo truck introduces significant hazards, including rollover risk due to high gross weight (20,000 lbs. or more) or a high center of gravity. Check the vehicle's data plate to determine if it is at a higher risk for rollover. Operators must mitigate risks associated with traffic density, heavy cargo loads, overhead clearances, and jerky starts, stops, or sharp turns.

4.3.1. Safety Equipment. Operators must utilize the DAF Form 1800 and appropriate safety equipment, including steel-toed boots, gloves (removing rings/jewelry before cargo handling), first aid kits, tire gauges, and fire extinguishers. If operating during hours of reduced visibility or on the flightline, reflective belts are required. Shovels, tow chains, cables, cold/rain weather gear, and materials for traction under wheels should be present for off-road operations. Hearing protection must be used if required.

4.3.2. Tire Changing Safety. If a tire change is required, move the vehicle to a firm, level surface away from traffic, bridges, or curves. Activate four-way flashers, block the wheels opposite the flat tire (e.g., block rear wheels if changing a front tire), and place the vehicle in park or low gear (if standard shift). If no park option exists, use neutral and engage the parking brake. Ensure the jack is rated for the vehicle's weight. Once lifted, operators must never get under the vehicle, and jack stands should be used if available. Before removing lug nuts, double-check that lug wedges are loose. Return the jack/lug wrench to the manufacturer's recommended location to prevent them from becoming projectiles and secure the damaged tire.

4.3.3. Hazardous Cargo & Foreign Object Damage (FOD). For transporting hazardous cargo, operators must refer to QTP 24-3 HAZMAT, *Explosives/Hazardous Materials Transport*. To prevent FOD, operators must remove debris from tires during daily inspections. Before entering the airfield, a physical check for loose/unsecured objects and a tire tread inspection must be accomplished (except for emergency response vehicles). Any vehicle driven on an unpaved surface must have a tire FOD inspection prior to re-entering the airfield. Vehicles frequenting the flightline must be equipped with a covered FOD container (emptied daily and marked "FOD") and a red/orange etched FOD picker (or with a red streamer attached), which must be annotated on the vehicle inspection form.

4.4. Vehicle Operation.

4.4.1. 4WD Operation (if equipped). For vehicles equipped with 4WD, the operator must utilize the transfer case appropriately based on terrain.

4.4.1.1. High Ratio, Two-Wheel Drive (H2): Used for standard street/highway driving. Provides best fuel economy. Can be shifted into/out of at any speed. Maintain a constant speed during the shift (avoid slowing down) and move the lever in one continuous motion.

4.4.1.2. High Ratio, Four-Wheel Drive (H4): Used for extra traction (snow, ice, light off-road, or snow plowing). Engages front axle. Can be shifted into/out of at any speed. Maintain a constant speed and move the lever in one continuous motion.

4.4.1.3. Neutral (N): Used strictly for towing. With the vehicle engine at idle, set the

parking brake, place the transmission in N, and shift the transfer case in one continuous motion.

4.4.1.4. Low Ratio, Four-Wheel Drive (L4): Used for deep sand, deep snow, or steep inclines/declines. Sends maximum power to all wheels. To engage L4, bring the vehicle to a complete stop. Place the transmission in Neutral (N), then shift the transfer case lever firmly into L4. Once engaged, shift the transmission back into a driving gear.

4.4.2. Refrigeration Unit Operation (if equipped). If operating a vehicle with a refrigeration unit, refer to the manufacturer's operator's manual. Adjust the thermostat, place the Cool – Idle – Heat switch to the idle position, and hold the pre-heat switch based on the ambient temperature (Above 60 degrees = 30 seconds; 32-60 degrees = 1 minute; 0-32 degrees = 2 minutes; Below 0 degrees = 3 minutes). Turn the start switch until the engine starts. If it fails, check for overheating or low oil pressure before retrying. Allow the engine to idle for two minutes before switching to cool or heat. Crucially, the operator must ensure the unit is entirely empty of all personnel prior to locking the doors.

Chapter 5

AIR BRAKES

5.1. Safe Operation of a CMV with an Air Brake System. Safe operation of a commercial motor vehicle (CMV) with an air brake system, requires knowledge of: Air Brake System nomenclature; the dangers of contaminated air supply; implications of severed or disconnected air lines between the power unit and the vehicle implications of low air pressure readings; procedures to conduct safe and accurate pre-trip inspections; procedures for conducting enroute and post-trip inspections of air actuated brake systems (ability to detect defects, which may cause the system to fail).

5.1.1. Air brakes are really three different braking systems:

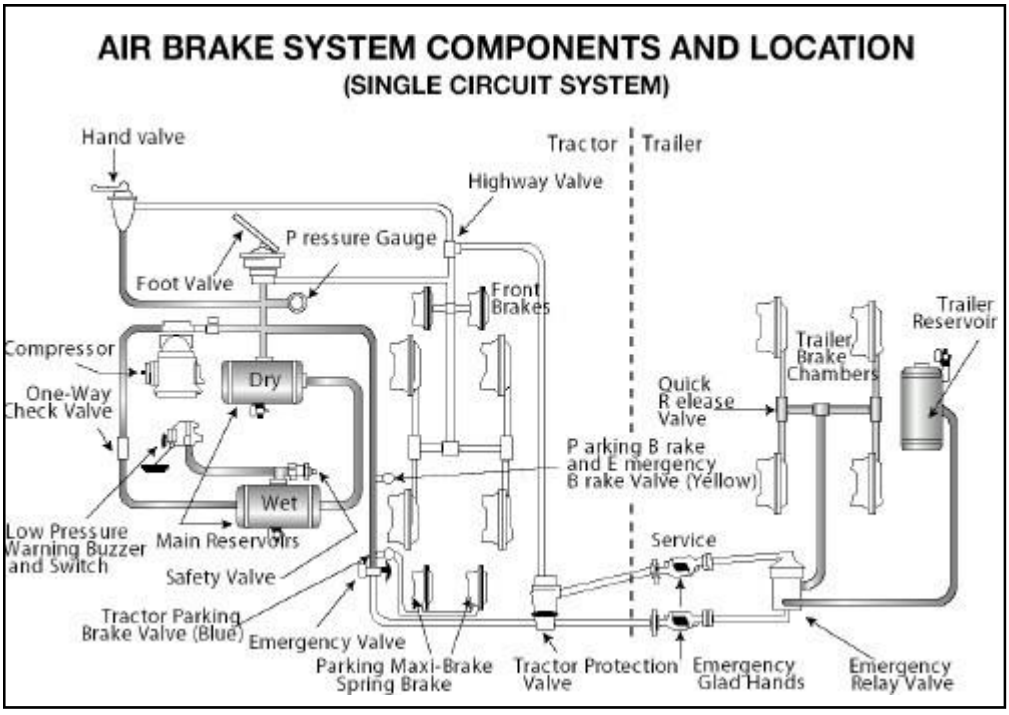
5.1.1.1. Service brake. Applies and releases the brakes when the operator uses the brake pedal during normal driving.

5.1.1.2. Parking brake. Applies and releases the parking brakes when the operator uses the parking brake control.

5.1.1.3. Emergency brake. Uses part of the service and parking brake systems to stop the vehicle in the event of a brake system failure.

5.1.2. Parts of an Air Brake System:

Figure 5.1. Air Brake System Components.



5.1.2.1. Air compressor. Pumps air into the air storage tanks (reservoirs). The air compressor is connected to the engine through gears or a v-belt. The compressor may be air cooled or may be cooled by the engine cooling system. It may have its own oil supply or be lubricated by engine oil. If the compressor has its own oil supply, check the oil level before driving.

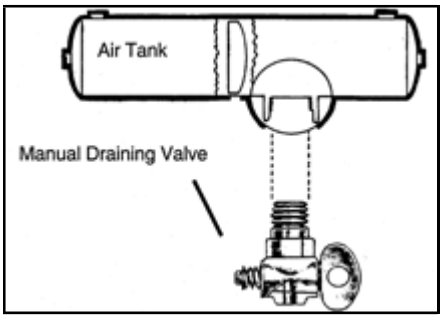
5.1.2.2. Air compressor governor. The governor controls when the air compressor will pump air into the air storage tanks. When air tank pressure rises to the "cut-out' level (around 125 pounds per square inch or psi), the governor stops the compressor from pumping air. When the tank pressure falls to the "cut-in" pressure (around 100 psi), the governor allows the compressor to start pumping again.

5.1.2.3. Air storage tanks. Air storage tanks are used to hold compressed air. The number and size of air tanks varies among vehicles. The tanks will hold enough air to allow the brakes to be used several times, even if the compressor stops working.

5.1.2.4. Air tank drains. Compressed air usually has some water and some compressor oil in it which is bad for the air brake system. The water can freeze in cold weather and cause brake failure. Drain the air tanks completely. Each air tank is equipped with a drain valve in the bottom. There are two types:

5.1.2.4.1. Manually operated. Turn a quarter turn or pull a cable to operate. See **Figure 5.2.** Manually drain the tank at the end of each day of driving.

Figure 5.2. Manual Drain Valve.



5.1.2.4.2. Automatic. The water and oil is automatically expelled. They may be equipped for manual draining as well. Automatic air tanks are available with electric heating devices to help prevent freezing of the automatic drain cold weather.

5.1.2.5. Alcohol evaporator. Some air brake systems have an alcohol evaporator to put alcohol into the air system. This helps to reduce the risk of ice in air brake valves and other parts during cold weather which can make the brakes stop working. Check and fill the alcohol container as necessary, every day during cold weather.

5.1.2.6. Safety valve. A safety relief valve is installed in the first tank the air compressor pumps air to. The safety valve protects the tank and the rest of the system from too much pressure. The valve is usually set to open at 150 psi. If the safety valve releases air,

something is wrong.

5.1.2.7. The brake pedal. Put on the brakes by pushing down the brake pedal. (It is also called the foot valve or treadle valve.)

5.1.2.7.1. Pushing the pedal down harder applies more air pressure. Letting up on the brake pedal reduces the air pressure and releases the brakes. Releasing the brakes releases some compressed air go out of the system, so the air pressure in the tanks is reduced.

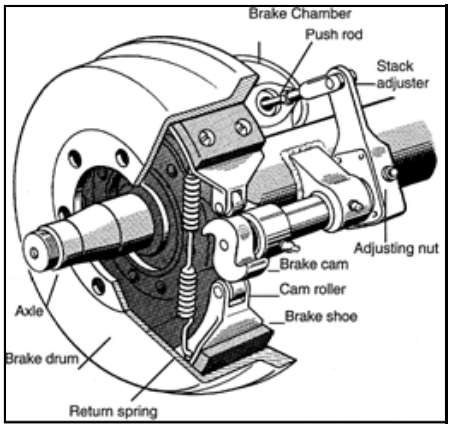
5.1.2.7.2. It must be made up by the air compressor. Pressing and releasing the pedal unnecessarily can release air out faster than the compressor can replace it. If the pressure gets too low, the brakes won't work.

5.1.3. Foundation brakes. Used at each wheel. The most common type is the s-cam drum brake, shown in **Figure 5.3.** Its parts include:

5.1.3.1. Brake drums, shoes, and linings. Located on each end of the vehicle's axles. The wheels are bolted to the drums. The braking mechanism is inside the drum. To stop, the brake shoes and linings are pushed against the inside of the drum. This causes friction, which slows the vehicle (and creates heat). The heat a drum can take without damage depends on how hard and how long the brakes are used. Too much heat can make the brakes stop working.

5.1.3.2. S-cam brakes. When the brake pedal is pushed, air is let into each brake chamber Air pressure pushes the rod out, moving the slack adjuster, thus twisting the brake camshaft. This turns the s-cam (so called because it is shaped like the letter "S"). The s-cam forces the brake shoes away from one another and presses them against the inside of the brake drum. When releasing the brake pedal, the s-cam rotates back and a spring pulls the brake shoes away from the drum, letting the wheels roll freely again.

Figure 5.3. S-cam Air Brakes.



5.1.3.3. Wedge brakes. The brake chamber push rod pushes a wedge directly between the ends of two brake shoes. This shoves them apart and against the inside of the brake drum. Wedge brakes may have a single brake chamber, or two brake chambers, pushing wedges

in at both ends of the brake shoes. Wedge type brakes may be self-adjusting or may require manual adjustment.

5.1.3.4. Disc brakes. In air-operated disc brakes, air pressure acts on a brake chamber and slack adjuster, like s-cam brakes. But instead of the s-cam, a "power screw" is used. The pressure of the brake chamber on the slack adjuster turns the power screw. The power screw clamps the disc or rotor between the brake lining pads of a caliper, similar to a large c-clamp.

5.1.4. Supply pressure gauges. All air-braked vehicles have a pressure gauge connected to the air tank. If the vehicle has a dual air brake system, there will be a gauge for each half of the system. (Or a single gauge with two needles.) Dual systems will be discussed later. These gauges tell the operator how much pressure is in the air tanks.

5.1.5. Application pressure gauge. This gauge shows how much air pressure is being applied to the brakes. (This gauge is not on all vehicles.) Increasing application pressure to hold the same speed means the brakes are fading. Slow down and use a lower gear. The need for increased pressure can also be caused by brakes out of adjustment, air leaks, or mechanical problems.

5.1.6. Low air pressure warning. A low air pressure warning signal is required on vehicles with air brakes. A warning signal that can be seen must come on before the air pressure in the tanks falls below 55 psi. (Or one half the compressor governor cutout pressure on older vehicles.) The warning is usually a red light. A buzzer may also come on.

5.1.6.1. Another type of warning is the "wig wag." This device drops a mechanical arm into the view when the pressure in the system drops below 55 psi. An automatic wig wag will rise out of the view when the pressure in the system goes above 55 psi. The manual reset type must be placed in the "out of view" position manually. It will not stay in place until the pressure in the system is above 55 psi.

5.1.7. Stop light switch. Drivers behind the vehicle must be warned when the brakes are put on. The air brake system does this with an electric switch that works by air pressure. The switch turns on the brake lights when the air brakes are put on.

5.1.8. Front brake limiting valve. Some older vehicles (made before 1975) have a front brake limiting valve and a control in the cab. The control is usually marked "normal" and "slippery." When the control is in the "slippery" position, the limiting valve cuts the "normal" air pressure to the front brakes by half.

5.1.8.1. Limiting valves were used to reduce the chance of the front wheels skidding on slippery surfaces. However, they actually reduce the stopping power of the vehicle.

5.1.8.2. Front wheel braking is good under all conditions. Tests have shown front wheel skids from braking are not likely even on ice. Make sure the control is in the "normal" position to have normal stopping power.

5.1.8.3. Many vehicles have automatic front wheel limiting valves. They reduce the air to

the front brakes except when the brakes are put on very hard (60 psi or more application pressure). These valves cannot be controlled by the driver.

5.1.9. Spring brakes. Tractor and straight truck spring brakes will come fully on when air pressure drops to a range of 20 to 45 psi (typically 20 to 30 psi). Do not wait for the brakes to come on automatically. When the low air pressure warning light and buzzer first come on, bring the vehicle to a safe stop right away, while the brakes can still be controlled. Brakes must be properly adjusted to work correctly and to prevent the braking power from being affected.

5.1.10. Parking brake controls. In newer vehicles with air brakes, engage the parking brakes using a diamond-shaped, yellow, push-pull control knob. Pull the knob out to put the parking brakes (spring brakes) on, and push it in to release them. On older vehicles, the parking brakes may be controlled by a lever. Use the parking brakes whenever the vehicle is parked. **Note:** Never push the brake pedal down when the spring brakes are on. The brakes could be damaged by the combined forces of the springs and the pressure.

5.1.10.1. Modulating control valves. A control handle on the dash board found in some vehicles that may be used to apply the spring brakes gradually. It is spring loaded so the operator has a feel for the braking action. The more the control lever is moved, the harder the spring brakes come on, so that the operator can control the spring brakes if the service brakes fail. When parking a vehicle with a modulating control valve, move the lever as far as it will go and hold it in place with the locking device.

5.1.10.2. Dual parking control valves. A dual valve is used when main air pressure is lost to release the spring brakes so the vehicle can be moved in an emergency. A push-pull type valve used to put the spring brakes for parking, and a spring loaded valve in the “out” position. When pushing the control in, air from the separate air tank releases the spring brakes so the vehicle can move. When releasing the button, the spring brakes come on again. There is only enough air in the separate tank to do this a few times. Plan carefully when moving or the vehicle may be stopped in a dangerous location when the separate air supply runs out.

5.1.10.3. Antilock Braking System. Truck tractors with air brakes built on or after March 1, 1997, and other air brakes vehicles, (trucks, buses, trailers, and converter dollies) built on or after March 1, 1998, are required to be equipped with antilock brakes. ABS is a computerized system that keeps the wheels from locking up during hard brake applications.

5.1.10.3.1. On newer vehicles, the malfunction lamp comes on at start-up for a bulb check, and then goes out quickly. On an older system, the lamp could stay on until the vehicle is driving over 5 mph.

5.1.10.3.2. If the lamp stays on after the bulb check, or goes on once the vehicle is under way, the operator may have lost ABS control at one or more wheels.

5.1.10.3.3. ABS is in addition to the normal brakes. It does not decrease or increase

the normal braking capability. It only activates when wheels are about to lock up.

5.1.11. Dual air brakes. Two separate air brake systems which use a single set of brake controls. Each system has its own air tanks, hoses, lines, etc. Both systems supply air to the trailer (if there is one).

5.1.11.1. One system (the "Primary") typically operates regular brakes on rear axle or axles.

5.1.11.2. The other system (the "Secondary") operates regular brakes on front axle (and possibly one rear axle).

5.1.11.3. Before driving a vehicle, allow time for the air compressor to build up a minimum of 100 psi pressure in both the primary and secondary systems, or to the recommended psi in the Manufacturer's Operator's Manual. **Note:** Pay attention to the low air pressure warning light and buzzer. Never operate if the warning light or buzzer is on (typically when the air pressure drops below 60 psi). Bring the vehicle to a stop right away and safely park the vehicle if this happens while driving.

5.1.12. Inspecting air brakes, there are added items to be included in Step 2 "Engine Compartment Checks", Step 5 "Walk-Around Inspecting", and Step 7 "Final Air Brake Check" of the Seven-Step Inspection Process. See **Attachment 4**. **Note:** If any of the tests fail, they must be reported to the VCO, the supervisor, and vehicle maintenance.

5.2. Using Air Brakes.

5.2.1. Normal stops. Push the brake pedal down. Control the pressure so the vehicle comes to a smooth, safe stop. If the vehicle has a manual transmission, do not push the clutch in until the engine rpm is down close to idle. When stopped, select a starting gear.

5.2.2. Emergency stops. When stopping in an emergency situation, the operator should brake in a way that will keep the vehicle in a straight line and allow the operator to turn if it becomes necessary. The operator can use controlled or stab braking methods. **Note:** If driving a vehicle with anti-lock brakes, read and follow the directions found in the owner's manual for stopping quickly:

5.2.2.1. Controlled braking:

5.2.2.1.1. Apply the brakes as hard as possible without locking the wheels.

5.2.2.1.2. Keep steering wheel movements very small while doing this.

5.2.2.1.3. Release the brakes if it is necessary to make a larger steering adjustment or if the wheels lock and reapply the brakes as soon as possible.

5.2.2.2. Stab braking:

5.2.2.2.1. Apply the brakes all the way.

5.2.2.2.2. Release brakes when wheels lock up.

5.2.2.2.3. As soon as the wheels start rolling, apply the brakes fully again.

5.2.3. Stopping distance. The total stopping distance for vehicles with air brake systems is made up of four different factors. Perception Distance plus Reaction Distance, plus Brake Lag Distance, plus Effective Braking Distance, equals the Total Stopping Distance.

5.2.4. Brake fading or failure:

5.2.4.1. Brakes can fade or fail from excessive heat caused by using them too much and not relying on the engine braking effect. Excessive heat causes chemical changes in the brake lining which reduces friction and causes expansion of the brake drums.

5.2.4.2. Brake fade is also affected by adjustment. When brakes are out of adjustment, they will not be able to share the workload with brakes that are properly adjusted, causing the other brakes to overheat and fade from overuse.

5.2.5. Proper Braking Technique. Once the vehicle is in the proper low gear, follow the proper braking technique: **Note:** The use of brakes on along and/or steep downgrade is only a supplement to the braking effect of the engine.

5.2.5.1. Apply the brakes just hard enough to feel a definite slowdown.

5.2.5.2. When the speed has been reduced to approximately 5 mph below the "safe" speed, release the brakes (for appx. 3 seconds).

5.2.5.3. When the speed has increased to the "safe" speed, repeat the above.

5.2.6. Low Air Pressure:

5.2.6.1. If the low air pressure warning comes on, stop and safely park the vehicle as soon as possible.

5.2.6.2. Spring brakes will engage when the air pressure drops into the range of 20 to 45 psi.

5.2.7. Parking Brakes. To use the parking brake, pull the parking brake control knob out. The control will be a yellow, diamond shaped knob labeled “parking brakes” on newer vehicles. On older vehicles, it may be a round blue knob or some other shape (including a lever that swings from side to side or up and down). **Note:** Never leave the vehicle unattended without applying the parking brakes or chock the wheels. The vehicle might roll away and cause injury and damage.

5.2.7.1. Do not use the parking brakes if the brakes are very hot or they can be damaged (e.g. from just coming down a steep grade). Let hot brakes cool before using the parking brakes.

5.2.7.2. If the vehicle does not have automatic air tank drains, drain air tanks to remove

the moisture and oil.

5.2.8. Safety Reminders.

5.2.8.1. ABS will not prevent power or turning skids.

5.2.8.2. ABS will not increase or decrease ultimate stopping power.

5.2.8.3. ABS will not necessarily shorten stopping distance.

5.2.8.4. ABS will not change the way the vehicle normally brakes.

Chapter 6

EXPLANATION AND DEMONSTRATION

6.1. Trainer Preparation and Safety Protocols. Prior to instruction, the trainer must establish a safe training location, reserve an appropriate vehicle, obtain the manufacturer's operator's manual, and ensure the trainee has completed a DAF Form 171. Both trainer and trainee must adhere strictly to safety protocols, including the removal of jewelry and identification tags, the use of wheel chocks when parked, and the mandatory use of seatbelts. The trainer should conduct an initial walk-around to familiarize themselves and the trainee with all warning labels, data plates (specifically noting load capacities and rollover risks), and vehicle dimensions.

6.1.1. Personal Protective Equipment and RM. Trainers and trainees must wear appropriate PPE, including safety steel-toed boots, gloves for cargo handling, and a reflective belt during hours of darkness or on the flightline. Throughout the demonstration, the trainer will actively model the five-step Air Force RM process, specifically assessing hazards related to heavy cargo loads, high centers of gravity, and overhead clearances.

6.2. Operator Maintenance and Vehicle Pre-Trip Inspection. The trainer will accomplish a comprehensive vehicle inspection alongside the trainee, documenting it on the DAF Form 1800. The trainer will demonstrate the mandated seven-step inspection method **(Attachment 4)** and utilize the inspection guide (**Attachment 2**). Crucially, the trainer will demonstrate how to conduct a proper tire Foreign Object Damage (FOD) check and, if the truck is equipped with air brakes, perform the comprehensive final in-cab air brake tests (testing the air leakage rate, the low-air pressure warning signal, and the emergency spring brake pop-out).

6.3. Operational Demonstration. Utilizing the vehicle data plate and the manufacturer's operator's manual, the trainer will explain the vehicle's capacities, controls, standard air pressure operating ranges, and all warning lights/buzzers. The trainer will demonstrate steering, turning, changing lanes, and navigating varying grades.

6.3.1. Cargo & Clearances: The trainer must emphasize oversized truck safety by demonstrating how to calculate overhead clearance, manage wide turning radiuses, and safely load, balance, and secure cargo using appropriate tie-downs and straps.

6.3.2. Specialized Systems: If equipped with air brakes, the trainer will demonstrate how to manage air pressure during stops and on downgrades using proper braking techniques. If equipped with 4WD, the trainer will demonstrate the proper shifting procedures for the transfer case (H2, H4, N, and L4) while in motion and stationery. If equipped with a refrigeration unit, the trainer will demonstrate the pre-heat and start-up procedures, emphasizing the safety check to ensure no personnel are inside before locking doors.

6.3.3. Backing Procedures. Due to the size and blind spots of an oversized cargo truck, Air Force regulations require the use of a spotter at all times when backing. The trainer will review standard hand signals and have the trainee serve as the spotter, demonstrating how to keep the spotter in sight in the mirrors at all times, and stopping the vehicle immediately if visual contact is lost.

6.4. Post-Operation Demonstration. Following the driving portion, the trainer will demonstrate after-operation procedures. This includes cleaning the vehicle, properly stowing cargo straps and chains, draining the air tanks (mandatory for air brake equipped vehicles to prevent moisture buildup), refueling, and executing the manufacturer's shut-down procedures. The trainer will park the vehicle in a level area, place the transmission in neutral (or park), apply the parking brake, and perform a post-operation walk-around, annotating any discrepancies on the DAF Form 1800.

Chapter 7

TRAINEE PERFORMANCE DEMONSTRATION AND EVALUATION

7.1. Trainee Performance and Practical Application. Following the demonstration, the trainee will transition to behind-the-wheel practice. The trainer must ensure safety at all times and immediately stop training if any safety infractions occur. If available, the vehicle should be marked with a "Driver in Training" sign. The trainee must don appropriate PPE (steel-toed boots, gloves, reflective belts), fasten their seatbelt, and properly adjust the air-ride seat and all mirrors to account for the truck's blind spots before moving.

7.2. Trainee Vehicle Inspection Execution. The trainee will conduct operator maintenance and pre/during inspections, utilizing **Attachment 2** as a guide. The trainee must verbally explain the items being inspected to the trainer, including the execution of a FOD check. If the vehicle is equipped with air brakes, the trainee must flawlessly execute and explain the in-cab air brake system tests without trainer assistance, ensuring the DAF Form 1800 is properly documented.

7.3. Trainee Operational Execution. The trainer will establish a road course containing turns, intersections, and varying terrain. The trainee will practice operating the oversized cargo truck forward and backward until they show proficiency. The trainee must successfully explain the vehicle's load capacities, air brake system indicators and warnings (if equipped), and rollover mitigation techniques.

7.3.1. Operations: They will practice wide turns, lane changes, entering/exiting highways, and speed control on upgrades/downgrades (demonstrating proper air pressure management and engine breaking if equipped).

7.3.2. Cargo & Systems: The trainee must demonstrate the ability to properly load, balance, and secure cargo. If the vehicle is so equipped, the trainee must successfully demonstrate shifting through all 4WD ranges and operating the refrigeration unit safely.

7.3.3. Backing Practice. The trainee will practice backing maneuvers utilizing the trainer (or another available trainee) as a spotter. The trainee must strictly adhere to using hand signals, maintaining visual contact via the mirrors, and stopping immediately if the spotter disappears.

7.4. Performance Evaluation. Once proficient, the trainee will complete the official performance evaluation using the checklist found in **Attachment 3**. The trainer will review **Attachment 3** with the trainee and answer any final questions. During the evaluation, the trainer will ensure all safety, PPE, cargo securing, air brake protocols, and operator manual warnings are strictly observed. The trainee must complete the evaluation checklist with zero instructor assistance.

7.4.1. Retraining and Re-evaluation. If the trainee fails any portion of the evaluation, they will be issued a "No-Go." The trainer will critique the observed weaknesses and re-demonstrate the failed items. The trainee must re-perform the specific tasks until they show proficiency before an official re-evaluation is conducted.

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

DAFI 13-213, *Airfield Driving*, 19 November 2023

DAFI 24-301, *Ground Transportation*, 9 January 2026

QTP24-3-HAZMAT, *Explosives/Hazardous Materials Transport*

Commercial Driver's License Manual, American Association of Motor Vehicle Administrators, March 2025

Adopted Forms

DAF Form 171, *Request for Driver's Training and Addition to U.S. Government Driver's License*

DAF Form 847, *Recommendation for Change of Publication*

DAF Form 1800, *Operator's Inspection Guide and Trouble Report*

Abbreviations and Acronyms

4WD—Four Wheel Drive

AAMVA—American Association of Motor Vehicle Administrators

ABS—Antilock Brake System

AFIMSC—Air Force Installation Mission Support Center

CDL—Commercial Driver's License

CMV—Commercial Motor Vehicle

DAF—Department of the Air Force

DAFI—Department of the Air Force Instruction

FOD—Foreign Object Damage

H2—High Ratio, Two Wheel Drive

H4—High Ratio, Four Wheel Drive

HAZMAT—Hazardous Materials

IAW—In Accordance With

L4—Low Ratio, Four Wheel Drive

N—Neutral

QTP—Qualification Training Package

RM—Risk Management

TO—Technical Order

VCO—Vehicle Control Official

Attachment 2

OVERSIZED CARGO TRUCK INSPECTION GUIDE

A2.1. Inspection Guide – Oversized Cargo Truck.

Inspection Guide - Oversized Cargo Truck

STEP 1. Vehicle Overview

- ☐ **Paperwork**
  - DAF Form 1800
- ☐ **Vehicle Approach**
  - Damage
  - Vehicle Leaning
  - Fluid Leaks (puddling on ground/frame)
  - Surrounding Hazards

STEP 2. Engine Compartment

- ☐ **Leaks/hoses/Electrical Wiring Insulation**
- ☐ **Oil Level**
- ☐ **Coolant Level**
- ☐ **Power Steering Fluid**
- ☐ **Windshield Washer Fluid**
- ☐ **Battery Connections & Tie-downs**
- ☐ **Automatic Transmission Fluid Level**
- ☐ **Engine Compartment Belts**

STEP 3. Engine Start/ Cab Check (Left/Front/Right)

- ☐ **Safe Start**
- ☐ **Gauges**
  - Oil Pressure Gauge
  - Air Pressure Gauge
  - Temperature Gauge (Coolant/Engine Oil)
  - Ammeter/Voltmeter
- ☐ **Warning Lights & Buzzers**
- ☐ **Mirrors & Windshield**
- ☐ **Wipers/Washers**
- ☐ **Emergency & Safety Equipment**
  - Red Reflective Triangles
  - Properly Charged & Rated Fire Extinguisher
  - Optional (Chains/Tire Changing Equip, Emergency Phone List)
- ☐ **Lights/Reflectors/Reflector Tape**
  - Dash Indicators for:
    - Left Turn Signal
    - Right Turn Signal
    - Four-Way Emergency Flashers
    - High Beam Headlight
    - ABS Indicator (If equipped)
    - Clearance Lights
  - Reflective Clean/Functional Light/Reflector Checks Include:
    - Headlights (clear)
    - Taillights (red)
    - Backing Lights (clear)
    - Turn Signals (amber, except in rear)
    - Four-Way Flashers (amber, except in rear)
    - Brake Lights (red)
    - Red Reflectors & Amber Reflectors (side/rear)
    - Reflective Tape Condition
- ☐ **Horn**
- ☐ **Heater/Defroster**
- ☐ **Brakes**
  - Parking Brake Check
  - Hydraulic Brake Check
  - Air Brake Check (if equipped)
  - Service Brake Check
- ☐ **Safety Belt**



STEP 4. Walk-Around Inspection

- ☐ **Steering**
  - Steering Box/Hoses/linkages
- ☐ **Suspension**
  - Springs/Air/Torque/Mounts
  - Shock Absorbers
- ☐ **Brakes**
  - Slack Adjustor's & Push-rods
  - Brake Chambers
  - Brake Hoses/Lines
  - Drum Brake/Brake Linings
- ☐ **Wheels**
  - Rims
  - Tires
  - Hub Oil Seals/Axle Seals
  - Lug Nuts
  - Doors
  - Mirrors
  - Fuel Tank

Attachment 3

PERFORMANCE TEST

A3.1. Instructions. Before beginning the performance test, the trainer will brief the trainee on the scenario the trainee will need to accomplish. He/she will be given additional directions and instructions as listed on the performance test checklist.

A3.2. Scoring.

A3.2.1. The trainer examiner will be scoring on oversized cargo truck operations and also the general safe driving practices. The examiner will give directions and instructions to the trainee in sufficient time for to execute a driving maneuver. He/she will not be asked to drive in an unsafe manner.

A3.2.2. The examiner will be making various marks on the performance test checklist. This does not necessarily mean the trainee have done anything wrong. It is in the best interest to concentrate on the operation of the oversized cargo truck. The trainer will explain the test results to the trainee at the conclusion of the performance test.

A3.2.3. Tasks being graded are listed on the following page; the trainee will be required to successfully pass all items.

A3.2.4. The instructor will stop the test at any time safe oversized cargo truck operations are not being followed or as deemed necessary for safety concerns.

Figure A3.1. Performance Test Checklist.

PERFORMANCE TEST CHECKLIST

Instructions to Certifier and Trainee:

Pre-Test Briefing: The Certifier will review this entire checklist with the trainee and brief the intended road course route. The trainee must understand that all driving directions will be provided in sufficient time to execute maneuvers safely.

Scoring Awareness: The Certifier will make various marks on this checklist during the evaluation. These marks do not inherently indicate a "No-Go"; the trainee must focus exclusively on safe vehicle operation.

Conduct of Test: No Trainer or Certifier assists are permitted during the evaluation. Results will be explained to the trainee upon the conclusion of the test.

Safety: The Certifier will stop the test immediately if unsafe operations occur or health concerns arise.

SECTION I. TRAINEE AND CERTIFIER INFORMATION

1.Trainee Name (Last, First MI)	**2. Unit/Office Symbol**
3. Certifier Name (Last, First MI)	**4. Unit/Office Symbol**

SECTION II. PERFORMANCE TEST INFORMATION

5. Vehicle Management Code for Vehicle Tested	**6. Date**

SECTION III. SCORESHEET

7. Equipment & Posture	Go	No Go
Verifies vehicle required forms & personal protective equipment	☐	☐

8. Pre-Operational & General Safety:	Go	No Go
Conducts 360 degree safety inspection when approaching veh	☐	☐
Inspects items identified on DAF1800, documents discrepancies	☐	☐
If equipped, Air Brake Check	☐	☐
Conducts ops check of controls & instruments	☐	☐
Sign DAF1800	☐	☐
Performs post trip inspection,reports any discrepancies to Veh Mgt	☐	☐

9. On-Road Operational Compliance:	Go	No Go
Obeys all traffic signs, signals, local/military traffic laws.	☐	☐
Completes course without accidents/moving violations.	☐	☐
Regular traffic/mirror checks/safe following distances.	☐	☐
Uses proper signals/smooth, safe lane changes.	☐	☐

10. Maneuvers (Turns and Curves):	Go	No Go
Turns: Checks traffic in all directions/signals properly.	☐	☐
Turns: Decelerates smoothly/proper gear for the turn.	☐	☐
Turns: Checks mirrors/avoids drifting into oncoming traffic.	☐	☐
Curves: Reduces speed/proper gear prior to entry	☐	☐
Curves: Maintains lane positioning/360-degree traffic checks	☐	☐

11. Intersections and Hazards:	Go	No Go
Int.: Decelerates gently/brakes smoothly on approach.	☐	☐
Int: Stop (no coasting) behind stop lines/signs/sidewalks.	☐	☐
R/R: Decelerates and shifts gears before the crossing.	☐	☐
R/R: Does not stop, shift, or change lanes while crossing.	☐	☐
Signs: Identifies height clearance signs on route	☐	☐

12. General Vehicle Control and Cargo Safety:	Go	No Go
Understands GVW, weight balance, and effects of overloading	☐	☐
Uses seat belts; follows safe loading/unloading and light protocols	☐	☐
Verifies surroundings; uses mirrors and spotters; controls speed	☐	☐
Monitors cargo/cargo area for shifting load or sign of	☐	☐

13. Remarks:

SECTION IV. CERTIFIERS ENDORSEMENT

I, the undersigned, do attest and certify that the trainee named herein has successfully completed all required training and is deemed proficient in the operation of the vehicle(s)/equipment listed in **Section II**. This certification is made to the best of my knowledge and belief, based on training conducted pursuant to the approved qualification training package.

14. Signature	**15. Date**

Attachment 4

SEVEN-STEP INSPECTION GUIDE

Figure A4.1. Seven-Step Inspection Guide.

SEVEN-STEP VEHICLE INSPECTION GUIDE

Step 1: Vehicle Overview

Review Last Vehicle Inspection Report. Drivers may have to make a vehicle inspection report in writing each day. The motor carrier must repair any items in the report that affect safety and certify on the report that repairs were made or were unnecessary. Only sign the previous report after verifying that any noted defects have been repaired or certified as not needing repair..

Step 2: Check Engine Compartment

Parking Brakes Are On and/or Wheels Chocked.

You may have to raise the hood, tilt the cab (ssecure any loose items to prevent them from falling and causing damage), or open the engine compartment door.

Check the following:

- Engine oil level.
- Coolant level in radiator; condition of hoses.

Check That the

- Power steering fluid level; hose condition (if so equipped).
- Windshield washer fluid level

Battery fluid level, connections, and tie downs (battery may be located elsewhere)

- Automatic transmission fluid level (may require engine to be running).
- Check belts for tightness and excessive wear (alternator, water pump, air compressor) --learn how much "give" the belts should have when adjusted right and check each one.
- Leaks in the engine compartment (fuel, coolant, oil, power steering fluid, hydraulic fluid, battery fluid).
- Cracked, worn electrical wiring insulation.
- Lower and secure hood, cab, or engine compartment door.

Step 3: Start Engine and Inspect Inside the Cab

Get In and Start Engine

Make sure parking brake is on.

Put gearshift neutral (or "park" if automatic).

Start engine; listen for unusual noises.

If equipped, check the Anti-lock Braking System (ABS) indicator lights. Light on dash should come on and then turn off. If it stays on the ABS is not working properly. For trailers only, if the yellow light on the left rear of the trailer stays on, the ABS is not working properly.

Look at the Gauges

Oil pressure. Should come up to normal within seconds after engine is started. See Figure 2.5

Air pressure. Pressure should build from 50 to 90 psi within 3 minutes. Build air pressure to governor cut-out (usually around 120 – 140 psi. **Know your vehicle's requirements.**

Ammeter and/or voltmeter. Should be in normal range(s).

Coolant temperature. Should begin gradual rise to normal operating range.

Engine oil temperature. Should begin gradual rise to normal operating range.

Warning lights and buzzers. Oil, coolant, charging circuit warning, and antilock brake system lights should go out right away.

Check Condition of Controls. Check all of the following for looseness, sticking, damage, or improper setting:

- Steering wheel.
- Clutch.
- Accelerator ("gas pedal").
- Brake controls.
- Foot brake.
- Trailer brake (if vehicle has one).
- Parking brake.
- Retarder controls (if vehicle has them).
- Transmission controls.
- Interaxle differential lock (if vehicle has one).
- Horn(s).
- Windshield wiper/washer.
- Lights.
- Headlights.
- Dimmer switch.
- Turn signal.
- Four-way flashers.
- Parking, clearance, identification, marker switch(es).

OIL PRESSURE

- Idling 5-20 PSI
- Operating 35-75 PSI
- Low, Dropping, Fluctuating: **STOP IMMEDIATELY!** Without oil the engine can be destroyed rapidly

Figure 2.5

Check Mirrors and Windshield. Inspect mirrors and windshield for cracks, dirt, illegal stickers, or other obstructions to seeing clearly. Clean and adjust as necessary.

Check Emergency Equipment

Check for safety equipment:

Spare electrical fuses (unless vehicles have circuit breakers).

Three red reflective triangles, 6 fuses or 3 liquid burning flares.

Properly charged and rated fire extinguisher.

Check for optional items such as:

- Chains (where winter conditions require).
- Tire changing equipment.
- List of emergency phone numbers
- Accident reporting kit (packet).

Check Safety Belt. Check that the safety belt is securely mounted, adjusts; latches properly and is not ripped or frayed.

Step 4: Turn Off Engine and Check Lights

Make sure the parking brake is set, turn off the engine, and take the key with you. Turn on headlights (low beams) and four-way emergency flashers and get out of the vehicle.

Step 5: Do Walk-around Inspection

- Go to front of vehicle and check that low beams are on, and both of the four-way flashers are working.
- Push dimmer switch and check that high beams work.
- Turn off headlights and four-way emergency flashers.
- Turn on parking, clearance, side-marker, and identification lights.
- Turn on right turn signal and start walk-around inspection.

General

- Walk around and inspect.
- Clean all lights, reflectors, and glass as you go along.

Left Front Side

- Driver's door glass should be clean.
- Door latches or locks should work properly.
- Left front wheel.
- Condition of wheel and rim--missing, bent, broken studs, clamps, lugs, or any signs of misalignment.
- Condition of tires--properly inflated, valve stem and cap OK, no serious cuts, bulges, or tread wear.
- Use wrench to test rust-streaked lug nuts, indicating looseness.
- Hub oil level OK, no leaks.
- Left front suspension.
- Condition of spring, spring hangers, shackles, U-bolts.
- Shock absorber condition.
- Left front brake.
- Condition of brake drum or disc.
- Condition of hoses.

Front

- Condition of front axle and steering system.
- No loose, worn, bent, damaged or missing parts.
- Must grab steering mechanism to test for looseness.
- Condition of windshield.
- Check for damage and clean if dirty.
- Check windshield wiper arms for proper spring tension.
- Check wiper blades for damage, "stiff" rubber, and securement.
- Lights and reflectors.
- Parking, clearance, and identification lights clean, operating, and proper color (amber at front).
- Reflectors clean and proper color (amber at front).
- Right front turn signal light clean, operating, and proper color (amber or white on signals facing forward).

Right Side

- Right front: check all items as done on left front.
- Primary and secondary safety cab locks engaged (if cab-over-engine design).
- Right fuel tank(s).
- Securely mounted, not damaged, or leaking.
- Fuel crossover line secure.
- Tank(s) contain enough fuel.
- Cap(s) on and secure.
- Condition of visible parts.
- Rear of engine--not leaking.
- Transmission--not leaking.
- Exhaust system--secure, not leaking, not touching wires, fuel, or air-lines.
- Frame and cross members--no bends or cracks.
- Air-lines and electrical wiring--secured against snagging, rubbing, wearing.
- Spare tire carrier or rack not damaged (if so equipped).
- Spare tire and/or wheel securely mounted.
- Spare tire and wheel adequate (proper size, properly inflated).
- Cargo securement (trucks).
- Cargo properly blocked, braced, tied, chained, etc.
- Header board adequate, secure
- Side boards, stakes strong enough, free of damage, properly set in place
- Canvas or tarp properly secured to prevent tearing, billowing, or blocking of mirrors.
- If oversize, all required signs (flags, lamps, and reflectors) safely and properly mounted and all required permits in driver's possession.
- Curbside cargo compartment doors in good condition, securely closed, latched/locked and required security seals in place.

Right Rear

- Condition of wheels and rims--no missing, bent, or broken spacers, studs, clamps, or lugs.
- Condition of tires--properly inflated, valve stems and caps OK, no serious cuts, bulges, tread wear, tires not rubbing each other, and nothing stuck between them.
- Tires same type, e.g., not mixed radial and bias types.
- Tires evenly matched (same sizes).
- Wheel bearing/seals not leaking.
- Suspension.
- Condition of spring(s), spring hangers, shackles, and U-bolts.
- Axle secure.
- Powered axle(s) not leaking lube (gear oil).
- Condition of torque rod arms, bushings.
- Condition of shock absorber(s).
- If retractable axle equipped, check condition of lift mechanism. If air powered, check for leaks.
- Condition of air ride components.
- Brakes.
- Brake adjustment.
- Condition of brake drum(s) or discs.
- Condition of hoses--look for any wear due to rubbing.
- Lights and reflectors.
- Side-marker lights clean, operating, and proper color (red at rear, others amber).
- Side-marker reflectors clean and proper color (red at rear, others amber).

Rear

- Lights and reflectors.
- Rear clearance and identification lights clean, operating, and proper color (red at rear).
- Reflectors clean and proper color (red at rear).
- Taillights clean, operating, and proper color (red at rear).
- Right rear turn signal operating, and proper color (red, yellow, or amber at rear).
- License plate(s) present, clean, and secured.
- Splash guards present, not damaged, properly fastened, not dragging on ground, or rubbing tires.
- Cargo secure (trucks).
- Cargo properly blocked, braced, tied, chained, etc.
- Tailboards up and properly secured.
- End gates free of damage, properly secured in stake sockets.
- Canvas or tarp (if required) properly secured to prevent tearing, billowing, or blocking of either the rearview mirrors or rear lights.
- If over-length, or over-width, make sure all signs and/or additional lights/flags are safely and properly mounted and all required permits are in driver's possession.
- Rear doors closed, latched/locked.

Left Side

- Check all items as done on right side, plus:
- Battery(ies)
- Battery box(es) securely mounted.
- Box has secure cover.
- Battery(ies) secured against movement.
- Battery(ies) not broken or leaking.
- Fluid in battery(ies) at proper level (except maintenance-free type).
- Cell caps present and securely tightened (except maintenance-free type).
- Vents in cell caps free of foreign material (except maintenance-free type).

Step 6: Check Signal Lights

- Get In and Turn Off Lights
- Turn off all lights.
- Turn on stop lights (apply trailer hand brake or have a helper put on the brake pedal).
- Turn on left turn signal lights.
- Get Out and Check Lights
- Left front turn signal light clean, operating and proper color (amber or white on signals facing the front).
- Left rear turn signal light and both stop lights clean, operating, and proper color (red, yellow, or amber).

Get In Vehicle

- Turn off lights not needed for driving.
- Check for all required papers, trip manifests, permits, etc.
- Secure all loose articles in cab (they might interfere with operation of the controls or hit you in a crash).
- Start the engine.

Step 7: Start the Engine and Check

Test for Hydraulic Leaks. If the vehicle has hydraulic brakes, pump the brake pedal three times. Then apply firm pressure to the pedal and hold for five seconds. The pedal should not move.

If it does, there may be a leak or other problem. Get it fixed before driving. If the vehicle has air brakes, do the checks described in Sections 5 and 6 of this manual.

Brake System

Test Parking Brake(s)

- Fasten safety belt
- Set parking brake (power unit only).
- Release trailer parking brake (if applicable).
- Place vehicle into a low gear.
- Gently pull forward against parking brake to make sure the parking brake holds.
- Repeat the same steps for the trailer with trailer parking brake set and power unit parking brakes released (if applicable).
- If it doesn't hold vehicle, it is faulty; get it fixed.

Test Service Brake Stopping Action

- Go about five miles per hour.
- Push brake pedal firmly
- "Pulling" to one side or the other can mean brake trouble.
- Any unusual brake pedal "feel" or delayed stopping action can mean trouble.
- If any defects are found during the inspection, they must be repaired before operating the vehicle. Federal and state laws prohibit driving an unsafe vehicle.